

Krishna Chemudupati

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EDUCATION

University of Pennsylvania – *Vagelos Integrated Program in Energy Research* Philadelphia, PA
MSE Electrical Engineering (3.6/4.0); BSE Electrical Engineering, BA Physics (3.3/4.0) May 2027
Concentrations & Minors: Mixed-Signal/RFICs, System-on-a-Chip, Mathematics, CS, Data Science
Coursework: Analog, Digital, and Mixed Signal Circuits (Grad Courses), Semiconductor Memory (Grad), Computer Organization (Grad), Embedded Devices (Grad), SystemVerilog, Signal Processing, Electrodynamics, Photonics

TECHNICAL SKILLS

IC Design & Simulation: Cadence Virtuoso, Electric VLSI, LTspice, Ngspice, Xschem, Magic, Altium, MATLAB
Digital Design & Verification: SystemVerilog, Verilog, Verilator, cocotb, Surfer, Quartus Prime, ModelSim
Programming: Python, C, C++, CUDA, MATLAB, NumPy, Pandas, Matplotlib, Git
Protocols & Debug: SPI, I2C, UART, Oscilloscope, Logic Analyzer, Keithley Analyzer, ATmega328PB, Raspberry Pi

EXPERIENCE

Camera Hardware Systems Engineering Intern May 2026 – Present
Qualcomm Technologies, Inc. San Diego, CA

- Studied the Qualcomm camera platform from hardware through kernel and user-space software, developing an end-to-end understanding of the camera SoC architecture
- Analyzed proprietary camera framework across user-mode and kernel-mode components, focusing on sensor communication, imaging pipelines, and hardware abstraction
- Enabled hardware SoC timer and GPIO-based LED synchronization on Qualcomm chipsets to support frame-line timing requirements for customer camera use cases
- Developed configurable control interfaces using session parameters to dynamically enable hardware features and update GPIO timing behavior

Ferroelectric Compute-in-Memory Researcher Feb 2025 – Present
Jariwala Lab, University of Pennsylvania Philadelphia, PA

- Characterizing FeDs in MFM/MIFM structures (AlScN with HfO₂ interlayers) across varied Sc concentrations, substrates, and electrodes for selector-free crossbar arrays enabling neuromorphic compute-in-memory
- BEOL peripheral circuitry design in 40nm CMOS; pA-level sensing via TIA/comparator; built PCB probe cards
- Co-first author on FeD multilevel arrays and author on low-T FeDs (*Nature Electronics*, under review)
- Developed Python scripts for automated data extraction and regression analysis of device switching behavior

PROJECTS

Programmable GHz-Class Phase-Locked Loop Tapeout | *SKY130, Xschem, ngspice, Magic* May 2026 – Present

- Designing a programmable frequency-synthesizer PLL targeting 32 MHz to 2.048 GHz using a current-starved ring VCO, charge pump, loop filter, phase-frequency detector, and programmable divide-by-2ⁿ feedback network
- Developing a bandgap reference to bias circuitry and improve charge-pump current stability across PVT variations
- Performing transistor-level simulation, Kvco extraction, loop stability analysis, Monte Carlo and PVT verification, followed by matching-aware layout, parasitic extraction, DRC/LVS, and post-layout validation

Transistor-Level VLSI Design Suite | *Cadence Virtuoso (45nm), Electric VLSI (22nm)* Spring – Fall 2025

- Memories: 3T1C DRAM with decoder, sense amps, refresh controller (68ns write, 0.21ns read); 6T SRAM with precharge, write drivers, differential sense amps (137.5ps, 35.2μW)
- Arithmetic: 4x4 Dadda multiplier with Elmore-delay sizing (36% faster at 125ps, 2.5x lower energy at 50fJ); 8-bit RCA with K-map XOR2, tau-model sizing (43.6% faster, 123ps, 3.56fJ)
- 16:1 LUT from 2:1 MUX hierarchy; sweeps: 68.8% faster (117ps), 58.4% higher freq, 46.6% lower energy
- Verified via boundary tests, counter-driven address sweeps, exhaustive coverage (256 cases, 2⁴ LUT); scripted Python to extract and analyze Cadence simulations; validated sense amp thresholds, retention, worst-case delays

Wideband Transimpedance Amplifier | *Cadence Virtuoso, 45nm CMOS* Nov – Dec 2025

- Designed 8-stage TIA for optical receivers: CG input, CS gain, differential conversion, 4-stage diff chain, buffer
- Achieved 406 kΩ transimpedance (112 dBΩ), 300 MHz BW, 121 nA_{rms} noise, 0.6 Vpp swing, 1.38 mW
- 45nm process characterization (ID-VDS, Vth, Cgs/Cgd, fT); hand calcs validated against SPICE for all metrics
- Stability analysis with phase margin verification; 49-page report comparing hand calculations vs. simulation